Cell Formats and Styles

Introduction

Throughout these lectures, you'll be working through completing files using Microsoft Excel 2016. When you see text within quotes or apostrophes, you need only enter the text, not the punctuation marks.
We will start with some basics. Formatting can be applied to a cell or its contents (or even part of its contents) in Microsoft Excel. This formatting can be applied using the same normal text settings found in Microsoft Word, but also less commonly used formats such as text orientation (among others). In addition, Excel allows you to format numbers (example: how many decimal places) or dates (example: to display the month in a specific way such as *August, Aug* or *08/*). It's important to realize that by applying formatting you alter only how the text is displayed – not the actual contents of the cell.

Formatting a Cell

If you've ever used Excel, you should be familiar with many of the regular formatting buttons in the *Font* and *Alignment* groups found on the **Home** tab. Excel has other buttons in these groups, and even more in the *Numbers* group.

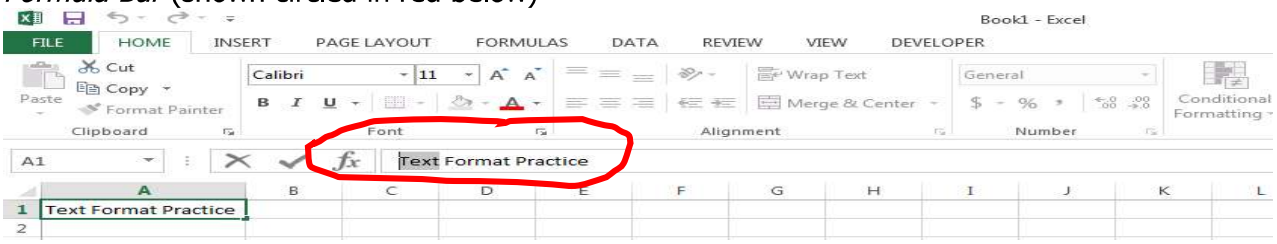
Formatting Text

Similar to using Microsoft Word, text can be formatted in Excel:

1. Open Excel.
2. Start with a new blank workbook (you can also press **[Ctrl-n]** to do this within an already open Excel application).
3. Type "Text Format Practice" in cell *A1* then press **[Ctrl-Enter]** (this allows you to enter the text but still remain in the same cell).
4. Try out the buttons in the *Font* group.

As you see, these actions effect the entire contents of the cell. You can also apply formatting to just a portion of the text instead of the whole cell.

5. Highlight the word(s) to be formatted (for your assignment, highlight "Text") in the *Formula Bar* (shown circled in red below)



6. Click on the required format button(s) – in this case, choose *Bold.*
7. Press **[Enter]** to see the result. Click on cell A1 again, and you can see that the formatting in the formula bar does not change, though the view in the cell does.

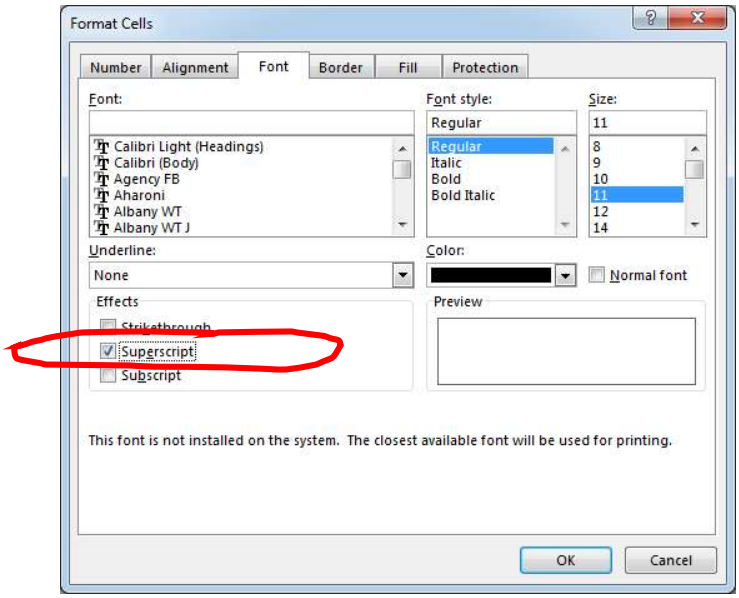
You can also apply formatting as you type:

8. Click in cell *A2,* then choose ***Symbol*** from the ***Font*** list by using the pull down.
9. Type **p = 3.14** then press **[Enter].** **Notice that if you go back to cell *A2*, the formula bar shows the letter p while the cell itself shows the pi symbol.

Tip: While entering text into a cell, you can use **[Ctrl-b]** for bold, [**Ctrl-i]** for italic, etc. in order to apply formatting as you type.

For superscript (or subscript):

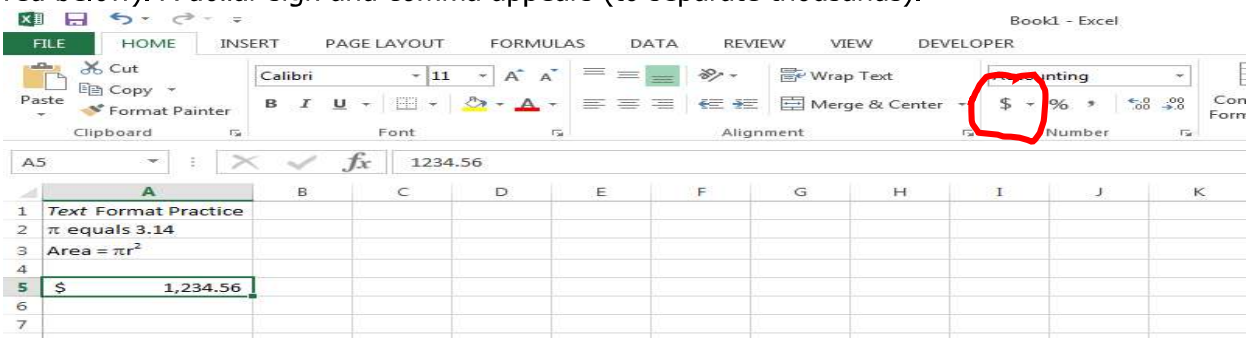
10. In cell *A3* type "Area ="
11. Change the ***Font*** to ***Symbol*** and type the letter **p**
12. Change the ***Font*** to ***Calibri*** and type the letter **r**
13. Next, click on the ***Font*** *group arrow* to display the *Format Cells* dialog box.
14. Under *Effects* turn on ***Superscript*** (as shown circled in red below) – click **[OK].**
15. Type **2,** then press **[Enter]** to finish typing.



Formatting Numbers

Excel provides special formatting for numbers and tries to give numeric data an appropriate format automatically when entered. Knowing how to change or clear these formats, as necessary, is part of the process of understanding these formats.

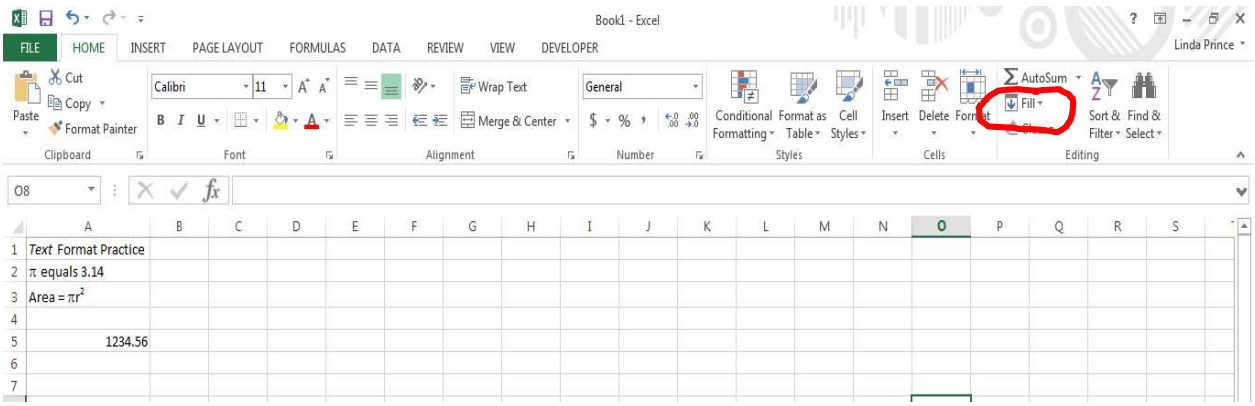
16. In your spreadsheet, click in cell A5.
17. Type "Number Format Practice" in in the cell then press **[Enter]** to move to cell *A6.*
18. Type "**1234.56"** and press **[Ctrl-Enter]** to remain in *A6.*
19. Click on the ***Accounting*** style button in the *Number* group (the $ as shown circled in red below). A dollar sign and comma appears (to separate thousands).



There are other buttons in the *Number* group – including within the **Number Format** pull down, such as Short or Long Date formats or Percentage. Also within the *Number* group are buttons such as **Decrease Decimal -** this will round the data to the nearest value within the displayed number of decimal places.

To clear formatting:

20. Click in cell A7 and type the word "rest", then press **[Ctrl-Enter]** to remain in *A7.*
21. Click on the **Clear** pull down button (shown circled in red below) in the *Editing* group and choose **Clear Formats.**



Note** You can either clear formats using the Clear pull down in the *Editing* group as described above, or by imposing one of the built in styles (**Accounting, Comma Style, etc.)** to override any existing formats.

Tip: Another way you can clear an unwanted format is to:

- click in a cell which hasn't been formatted
- click on the **Format Painter** button
- click in the cell with the format you want to clear.

You can very easily impose an unintended format on a cell. For example, if you type a dollar amount (with dollar sign) into a cell, then go back and type over it with other numbers, even if you do not type the dollar sign, it will apply the currency format.

Sometimes you can get very strange results using automatic formatting - you need to comprehend what is happening:

22. Click in cell *A7.*

23. Type **4-2** and press **[Ctrl-Enter]**.

Excel interprets this as a date (April 2nd in the current year). The cell view shows 2-Apr, while the formula bar shows the full date with slashes between.

If you wanted to calculate 4 minus 2, you'd need to put an equal sign before the calculation.

Note**Whether Excel interprets yours as a date depends on what operating system as well as which Excel version you are using – if yours does not give a date format, don't panic – just change the format to a date. ☺

Excel stores dates as numbers. As a matter of fact, dates in Excel are calculated as the number of days since January 1st, 1900.

24. In cell *A8,* type 5/1/16, then press **[Ctrl-Enter]** to stay in the cell.

25. Press the **DELETE** key.

26. Still in cell *A8,* type in the formula **=A7/8** and again press **[Ctrl-Enter]**.

In cell *A8*, the result shows as date because the cell has retained that original date format you deleted and also because cell A7 is a date format. The value of *A8* will change, but will still show as a date no matter what number you try to type into cell *A7*.

27. Click in cell *A7* and type the number 75. Notice that both A7 and A8 change, but both remain dates.

Numeric Text

Excel does not allow leading zeros in a numeric format. This makes it difficult, for example, to store social security numbers (*not that you would EVER store social security numbers in an Excel sheet*) for some people born or registered in New York (who may have SSNs that begin

with zero). The answer is to store the number as numeric *text*. To do this, precede the number with a single quote (or apostrophe):

28. Move down to cell *A9*

29. Type in a single quote (the key to the left of the **[Enter]** key) then a nine digit number beginning with zero.

30. When you press **[Enter],** the leading zero will be displayed and the single quote will not show.

31. Go back to A9 and click the $ in the *Number* group - you will find the display doesn't change other than to shift a bit.

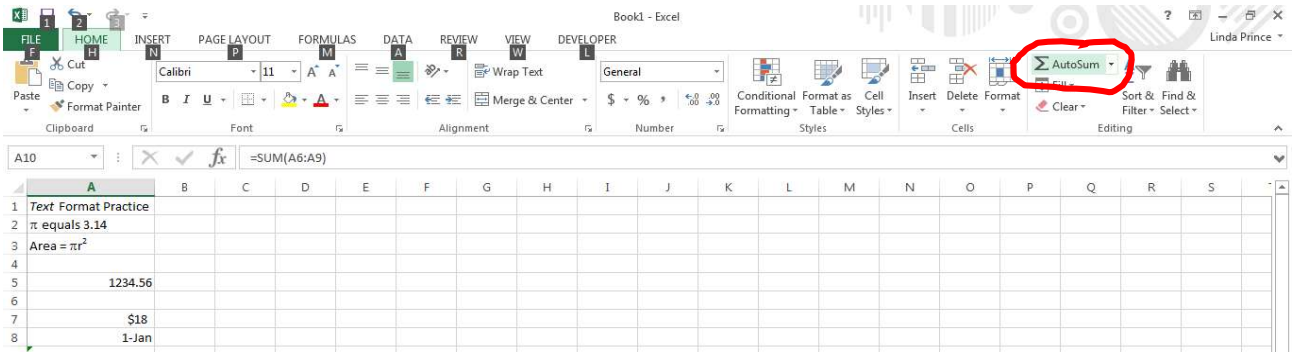
If you were to type in a number without the leading single quote, then numeric styles would apply when you clicked the $, but because the number is stored as text, Excel will not apply numeric styles. Entering text using a single quote is also very useful for other needs in Excel. Look further at these by watching the first lecture video.

- showing equations with a leading equal sign (so that the equation shows and doesn't calculate)
- showing numeric text which include dashes and/or slashes that would otherwise be interpreted as dates
- Showing numeric text which includes a colon that would otherwise be interpreted as a time
- showing a leading plus sign
- where you have heading with a numeric value which shouldn't be included in calculations

Try totaling the values in cells *A7* to *A9*:

32. Click in cell *A10.*

33. Click on the **AutoSum** button in the *Filtering* group (shown circled in red below).



34. Within the parentheses of the SUM formula, type the range *A7:A9* and press **[Enter]**.

35. Note that the answer doesn't include the social security number, nor is it a number!

36. Now type the number '63' into *A9* – the value of cell *A10* changes but the format stays.

37. Change the formula in cell *A10* so that the range is A6:A9.

Now the value of cell A10 changes, and so does the format. Also notice that the value does not match the total of only cells A6 and A9, even though those appear to be the only numeric values. Remember, dates are considered numeric to Excel, and so it adds them into the total as well.

38. Next, move to cell *A11* and enter another numeric text value that leads with a zero and is the length of a social security number.

Formatting Cells

Now that you've formatted the contents of cells, now we will format the cells themselves. Buttons within the Ribbon let you set formatting to a cell – you can justify the layout, choose the color that fills the cell or of the font within the cell, add borders, etc.

39. Press **[Ctrl-Home] –** this will move the cursor to cell *A1.*

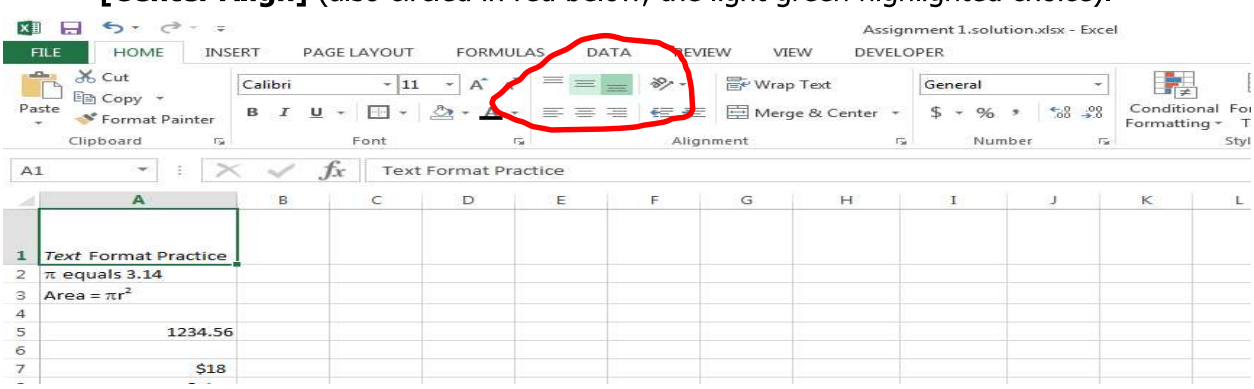
40. Use the pull down on the **[Font Color]** and move the cursor through the color choices. This time, the text in cell *A1* changes color as you move the cursor. Click on **[Automatic]** to leave the text color black.

41. Use the pull down on the **[Fill Color]** and move the cursor through the color choices. Cell *A1* fills with different colors behind the text as you move the cursor. Click on **[No Fill]** to leave the cell color blank.

Using the buttons in the *Alignment* group allow you to change the indentation of your text, justify its display or even alter the angle direction.

42. Move your cursor between the 1 and 2 labels on the rows until the cursor changes to a plus sign with arrows on the vertical axis. Holding the left mouse button, drag down to increase the row height to 45.00.

43. As you enlarge the row, you'll notice that the text moves with it and stays at the bottom of cell *A1.* The default alignment selection is **[Bottom Align]** (the dark green highlighted option shown circled in red below). To the left of the **[Bottom Align]** is **[Center Align]** (also circled in red below, the light green highlighted choice).



44. Click **[Center Align]** and notice that it works both vertically and horizontally.

45. Next, click on the **[Orientation]** pull down (circled in red above, it has an ab above a diagonal arrow) and choose **Angle Clockwise**. You should no longer be able to see the entire text – the row height of 45 that we selected earlier is smaller than the angled height of the text.
46. Click on **[Wrap Text]**, which is the button to the right of Orientation, also found in the *Alignment* group and look what happens.

Advanced Formatting

We've been using the Ribbon buttons to work with formatting cells. Now we'll look at more options usable within the *Format Cells* area.

47. Click in cell *B2* then click on either the *Font, Alignment* or *Number* group arrow – any of these arrows will open the *Format Cells* box. You can also open *Format Cells* by selecting **[Format Cell Alignment]** using the **[Orientation]** pull down.

The Number Tab

In the *Format Cells* area, click on the *Number* tab. This is where you can choose how you want a number to appear. The default is *General,* which has no specific number format and you'll see several alternate *Category* items that you can select to alter the number format:

48. ***Number Category*** – gives you to option to choose a number of decimal places, include a comma separator for thousands, and/or alter the display of negative numbers in both style and color.
49. ***Currency Category*** - gives you to option to choose a number of decimal places, include a symbol (such as a dollar sign or text currency values like EUR for euros), and/or alter the display of negative numbers in both style and color.
50. ***Accounting Category*** – gives you to option to choose a number of decimal places, include a symbol (such as a dollar sign or text currency values like EUR for euros). The notable difference between Accounting and Currency is the placement of the symbol. Accounting puts the symbol on the left side of the cell while Currency keeps it with the number.
51. Choose the **Text** format and click **[OK]**.
52. Type the number 488 into the Cell *B2.*
53. Press **[Enter].** Notice that the cell contains 488, but it is oriented left and there is a small green triangle in the upper left corner of the cell.

The Alignment Tab

54. Click in cell *B4* then click on either the *Font, Alignment* or *Number* group arrow – any of these arrows will open the *Format Cells* box.
55. Click on the *Alignment* tab.

56. Under *Text control* click the box to the left of **Shrink to fit** and click **[OK]**,
57. Type "Shrinking Practice" and press **[Enter]**. Note the size of the text.
58. Move your cursor between the B and C labels on the columns until the cursor changes to a plus sign with arrows on the horizontal axis. Holding the left mouse button, drag right to increase the column width to 13.00. You'll notice the text size increases as the column widens.

The Font Tab

59. Click in cell *B6* then click on either the *Font, Alignment* or *Number* group arrow – any of these arrows will open the *Format Cells* box.
60. Click on the *Font* tab.
61. Use the *Underline* pull down and select **Double Accounting**.
62. In the *Effects* area, click the box to the left of **Strikethrough.**
63. Click **[OK].**
64. Type "Font Formatting" and watch what happens both as you type and after you press **[Enter]**.

The Border Tab

65. Click in cell *B7* then click on either the *Font, Alignment* or *Number* group arrow – any of these arrows will open the *Format Cells* box.
66. Click on the *Border* tab. This is where you can choose border styles, colors and set up what areas of the cell you want to include borders.
67. Select the dotted line border *Style* directly below the None option.
68. Click on the pull down for *Color* and choose purple (the last in the row of Standard colors)
69. Click on the *Outline* choice under Presets and then also add the diagonals (below the diagram box on both the left and right.
70. Click **[OK]**. You should see an X in the cell as a border pattern.

You can apply borders to a group of cells at one time by selecting all the cells you want to include before making your border selections.

The Fill Tab

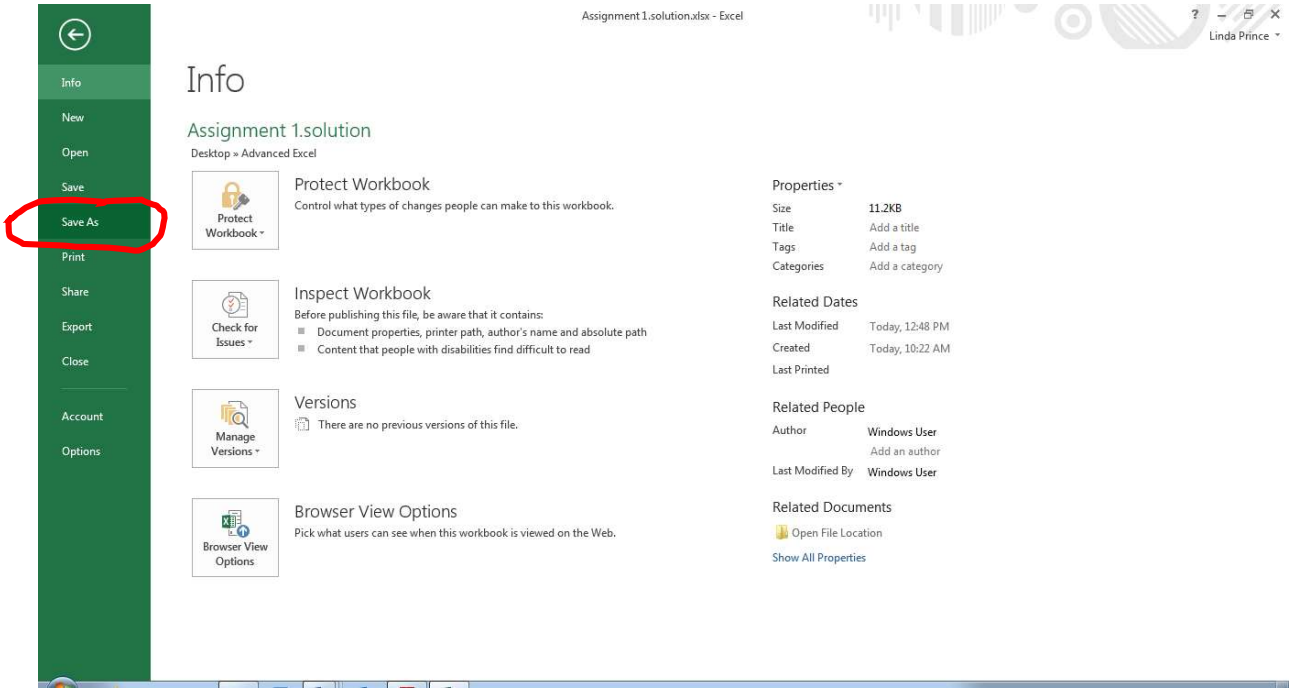
71. Click in cell *B9* then click on either the *Font, Alignment* or *Number* group arrow – any of these arrows will open the *Format Cells* box.
72. Click on the *Fill* tab. This is where you can choose background colors, patterns, pattern colors and effects.
73. Click on the **[Fill Effects]** button and choose a maroon color for color 1 and gold for color 2 in the *Colors* section.
74. Click on **[OK]** and then click **[OK]** again.
75. Click in cell *B11* then click on either the *Font, Alignment* or *Number* group arrow – any of these arrows will open the *Format Cells* box.

76. Click on the *Fill* tab.
77. Choose a green *Pattern Color* using the pull down.
78. Choose the Thin Horizontal Crosshatch *Pattern Style* (bottom row, second from right).
79. Click on **[OK]**.
80. Use the **[Format Painter]** to copy your pattern to cell *B12*

The Protection Tab

81. Click in cell *B14* then click on either the *Font, Alignment* or *Number* group arrow – any of these arrows will open the *Format Cells* box.
82. Click on the *Protection* tab. This is where you can lock cells or hide formulas. In order for either of these options to take effect, you have to protect the worksheet.
83. Check the *Hidden* box and click **[OK]**.
84. Type "=A6-15" into cell *B14*, then click **[Ctrl-Enter]** to remain in the cell.
85. Click on the *Review* tab and click the *Protect Sheet* option.
86. Leave the default options of protection and type '123' as a password.
87. You'll need to retype the 123 password to confirm.
88. Go back to the *Home* tab and notice that you can no longer select most of the *Font* choices, none of the *Alignment, Number or Styles* options and you also cannot select the arrows to open the *Format Cells* box. Also notice that the formula in cell *B14* does not appear in the *Formula Bar.*

Click on the *File* tab, then choose *Save As* (shown circled in red below). Save the file to your computer, then answer the cell formats assessment based on your file and/or comprehension of the lecture.



Conditional Formatting

Excel allows you to choose differing formats based on the contents of cells.

1. Open the SalesData.xlsx file found in Canvas.
2. Select column *E.*
3. Click on the ***Conditional Formatting*** pull down
4. Choose **Highlight Cell Rules** and choose **Less Than…**
5. In the *Less Than* box, type 30 in the format box and choose *Green Fill with Dark Green Text* from the 'with' pull down.
6. Click **[OK]**.

Conditional Formatting also allows you to color cells based on averages, scales or options you customize.

Look further at these by watching the second lecture video.

You can practice different conditional formatting if you choose in this file. You can also add your own rules by choosing *Manage Rules* in the Conditional Formatting pull down.

Close the Excel sheet without saving changes.